

Suleyman Hajizadeh

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COMPUTATIONAL BIOLOGIST & BIOINFORMATICIAN

Baku, Azerbaijan

RESEARCH INTERESTS

Computational genomics & transcriptomics · Structural bioinformatics & computational biophysics · Statistical machine learning applied to cancer biology · Reproducible pipeline engineering · Molecular evolution & phylogenomics

EDUCATION

Western Caspian University

Baku, Azerbaijan · 2022 – Jun 2026

Bachelor of Science in Biology

- ▶ Relevant coursework: Molecular Biology, Genetics, Biochemistry, Biostatistics, Computational Methods in Biology
- ▶ Independent research in computational cancer genomics conducted in parallel with degree coursework

RESEARCH & TECHNICAL PROJECTS

High-Throughput RNA-Seq Pipeline — Triple-Negative Breast Cancer

2024 – Present

Institute of Molecular Biology & Biotechnology (IMBB), Baku

- ▶ Developed a **Snakemake** workflow (SRA → FastQC → Trimmomatic → HISAT2/GRCh38 → featureCounts → DESeq2) for fully reproducible bulk RNA-seq analysis of GSE183947 (TNBC cohort)
- ▶ Implemented **DESeq2** differential expression with statistical commentary: derived negative binomial model assumptions, BH-FDR correction mathematics, and VST vs. rlog trade-offs in a dedicated *Statistical Methods* primer
- ▶ Designed a nested Leave-One-Out Cross-Validation classifier with feature selection inside training folds to prevent data leakage; permutation test empirical p-value < 0.05
- ▶ Manuscript: "*Machine Learning-Based Identification of Transcriptomic Subtypes and Key Biomarkers in Triple-Negative Breast Cancer*" — submitted to **PLOS ONE** (under review)

Computational Phylogenomics CLI — COL1A1 Vertebrate Evolution

2025

- ▶ Built `phylo_pipeline.py` — a **pure Python** command-line tool that fetches CDS sequences from NCBI Entrez, performs pairwise alignment, and computes **K2P** and **JC69 distance matrices from mathematical formulas**
- ▶ Implemented the **Neighbor-Joining algorithm from scratch** in NumPy; reconstructed the evolutionary history of COL1A1 across 7 vertebrate species (*H. sapiens*, *Pan troglodytes*, *M. musculus*, *R. norvegicus*, *B. taurus*, *G. gallus*, *D. rerio*)
- ▶ Outputs: Newick tree file (`.nwk`) and high-resolution cladogram PNG — **no MEGA GUI dependency**

Protein Computational Biophysics Toolkit2025

- Parsed crystallographic PDB coordinate files with Biopython; extracted Cα atoms and computed 3D Euclidean distance matrices and backbone RMSD between structural domains
- Implemented **hydrophobic core detection** using Kyte-Doolittle scale weighted by spatial packing density (8 Å threshold)
- Coded a **Miyazawa-Jernigan statistical contact potential** loop to estimate non-covalent folding free energies at residue-pair resolution
- Computed Pearson correlation between Cα packing density and crystallographic B-factors ($r = -0.37$ for *insulin 1COH*), mathematically validating core rigidity

AlphaFold2 Structural Analysis — AKT1 Kinase (TNBC)2024

- Predicted 3D structure of AKT1 hub kinase using ColabFold v1.6.1 (AlphaFold2 multimer); mean pLDDT = 65.61 across full chain
- Generated per-residue pLDDT confidence plots, Ramachandran backbone torsion analysis, Cα contact maps, and PyMOL publication-quality renders

TECHNICAL SKILLS

Languages	Python (Advanced) · R (Intermediate) · Bash/Shell scripting
Sci. Python	Biopython · NumPy · Pandas · SciPy · Scikit-Learn · Matplotlib
R / Bioconductor	DESeq2 · WGCNA · edgeR · pROC · ggplot2 · limma
Pipelines	Snakemake · Conda · Git/GitHub · SRA Toolkit · HISAT2 · featureCounts
Structural Bio	AlphaFold2 (ColabFold) · PyMOL · PDB parsing · RMSD · contact maps
Statistics / ML	Negative Binomial model · BH-FDR · Nested CV · Random Forest · PCA · GSEA
From Scratch	K2P distance matrix · JC69 model · NJ algorithm · MJ contact potential · hydrophobic core detection

PUBLICATIONS & MANUSCRIPTS

<i>Machine Learning-Based Identification of Transcriptomic Subtypes and Key Biomarkers in Triple-Negative Breast Cancer</i>	UNDER REVIEW
Hajizadeh S. · PLOS ONE · Submitted 2025	

AFFILIATIONS & TRAINING

Institute of Molecular Biology & Biotechnology (IMBB) <i>Bioinformatician (independent researcher)</i>	Baku, Azerbaijan · 2023 – Present
Data Analysis for Life Sciences	Harvard University · edX · Expected Jul 2026
Bioinformatics MicroMasters	University of Maryland Global Campus · edX · Expected Aug 2026

LANGUAGES

Azerbaijani

Native

English

Professional (IELTS target 7.5+, Jul 2026)

Russian

Conversational